Race, explain and transfer

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Evaluate a model against measured performance and defend design decisions.

SUCCESS CRITERIA - I can...

1. Report measured distance, time and average speed.
2. Compare actual speed with a stated prediction.
3. Use evidence to explain three physical design choices.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

At unchanged wheel rpm, circumference becomes 6 times as large. Compare ideal travel speed and state one real-world limitation.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: average speed = distance/time
relative deviation = |measured - predicted|/|predicted| × 100%. Prediction must be nonzero. Record real race conditions; this compares with a prediction, not symmetric percent difference.

Worked reasoning: A car covers 3 m in 2 s: average speed 1.5 m/s. A complete explanation also uses its circuit, gearing, friction and evidence from repeated runs; one race time does not isolate a cause.

Your turn: Draw an energy-flow diagram from battery to motion and other transfers; annotate it with one actual result, clearly labeled as measured.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: ① ② ③

03 Read and interpret the model

Draw or annotate your final car. Label three choices involving wheels, gears, alignment, wiring or mass, then connect each to a physics idea.



Marzano check - circle: ① ② ③

04 Apply the physics

A supplied car covers 5 m in 10 s; its predicted speed was 0.60 m/s. Complete the comparison table. Find measured speed and use the 0.10 m/s difference to calculate percent deviation relative to the prediction. State whether it was faster or slower.

Distance (m)	Time (s)	Measured v	Predicted v
___	___	___	___

Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

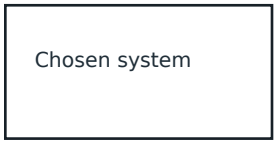
Use your actual race record to defend one successful design choice and identify one revision for a future build. Cite data or a specific observation for each.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A device receives 60 J mechanical energy at its shaft and produces electrical energy at its terminals. Identify it and name one possible loss.
Label input, useful output and other energy transfers.

Draw and label all energy transfers across the boundary.



Stored energy change: _____ J

Marzano check - circle: 1 2 3

07 Exit check

Choose one problem on this sheet. Circle your current 1, 2 or 3 self-check and name the next step that would improve your explanation.

Problem chosen	What I can explain	Next time
___	___	___

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Propose one redesign and predict two possible effects, including a tradeoff. State the measurement that would support or challenge your prediction.

Proposed change	Benefit / tradeoff	Test
___	___	___
___	___	___

Marzano check - circle: 1 2 3